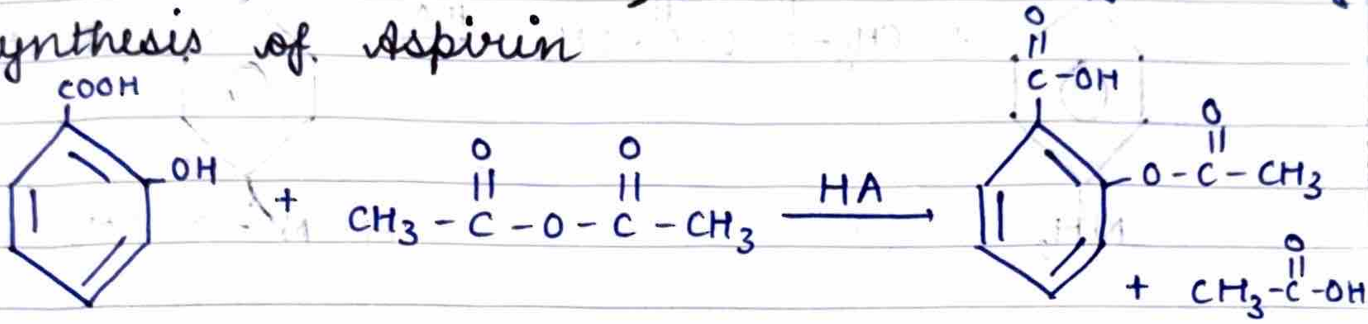
Applications
- Pain relief
- Anti inflammatory.

Unit → 2

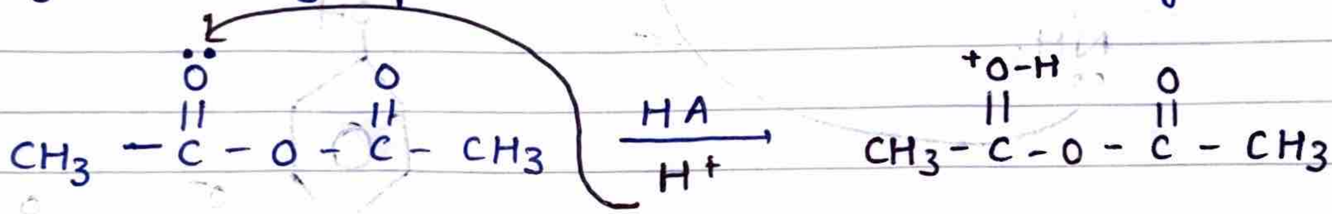
→ Synthesis of Aspirin



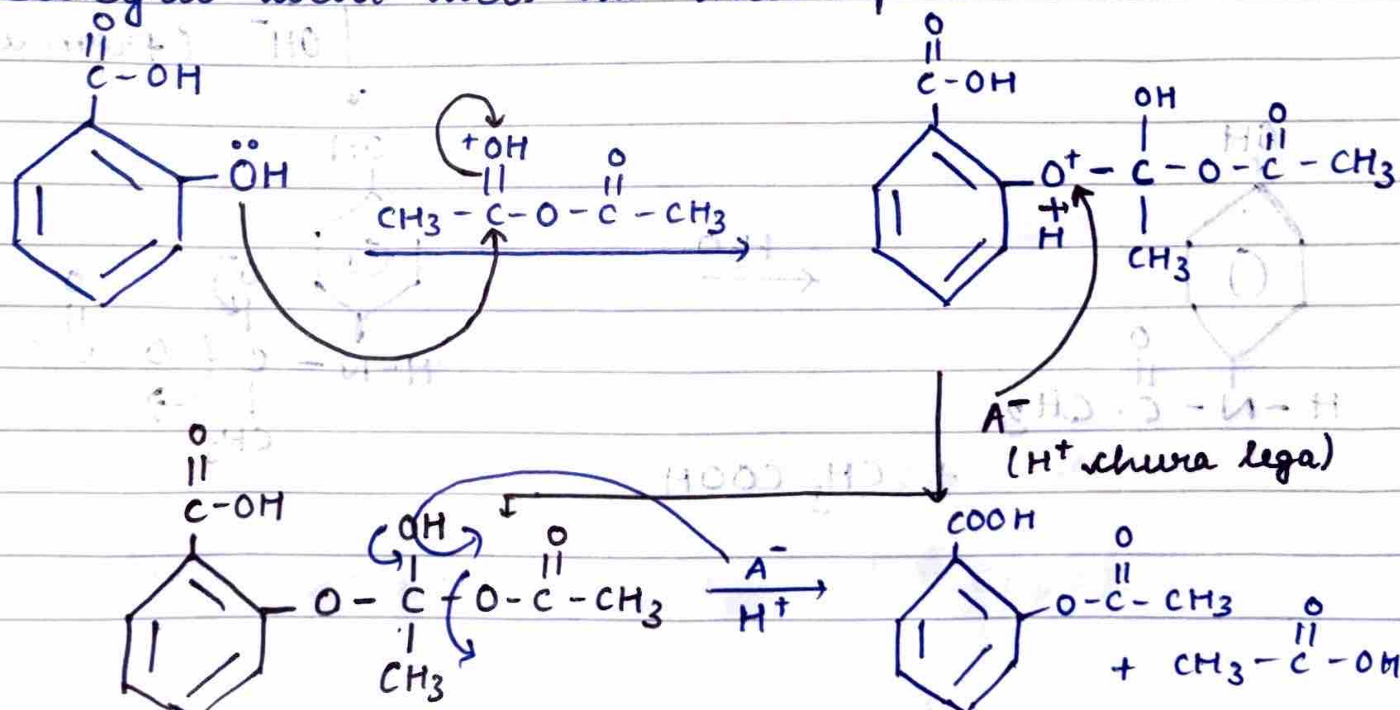
Salicylic acid combines with acetic anhydride to form aspirin and acetic acid.

- Mechanism

1. Protonation of acetic anhydride
$CH_3COOOCCH_3$ pe HA ka H^+ attack karega



2. Addition of nucleophile to this formed complex
Salicylic acid acts as nucleophile



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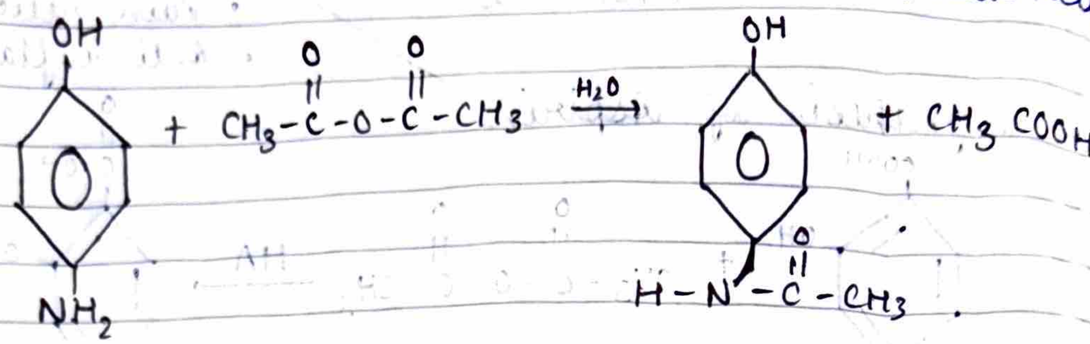
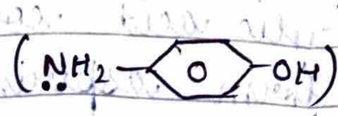
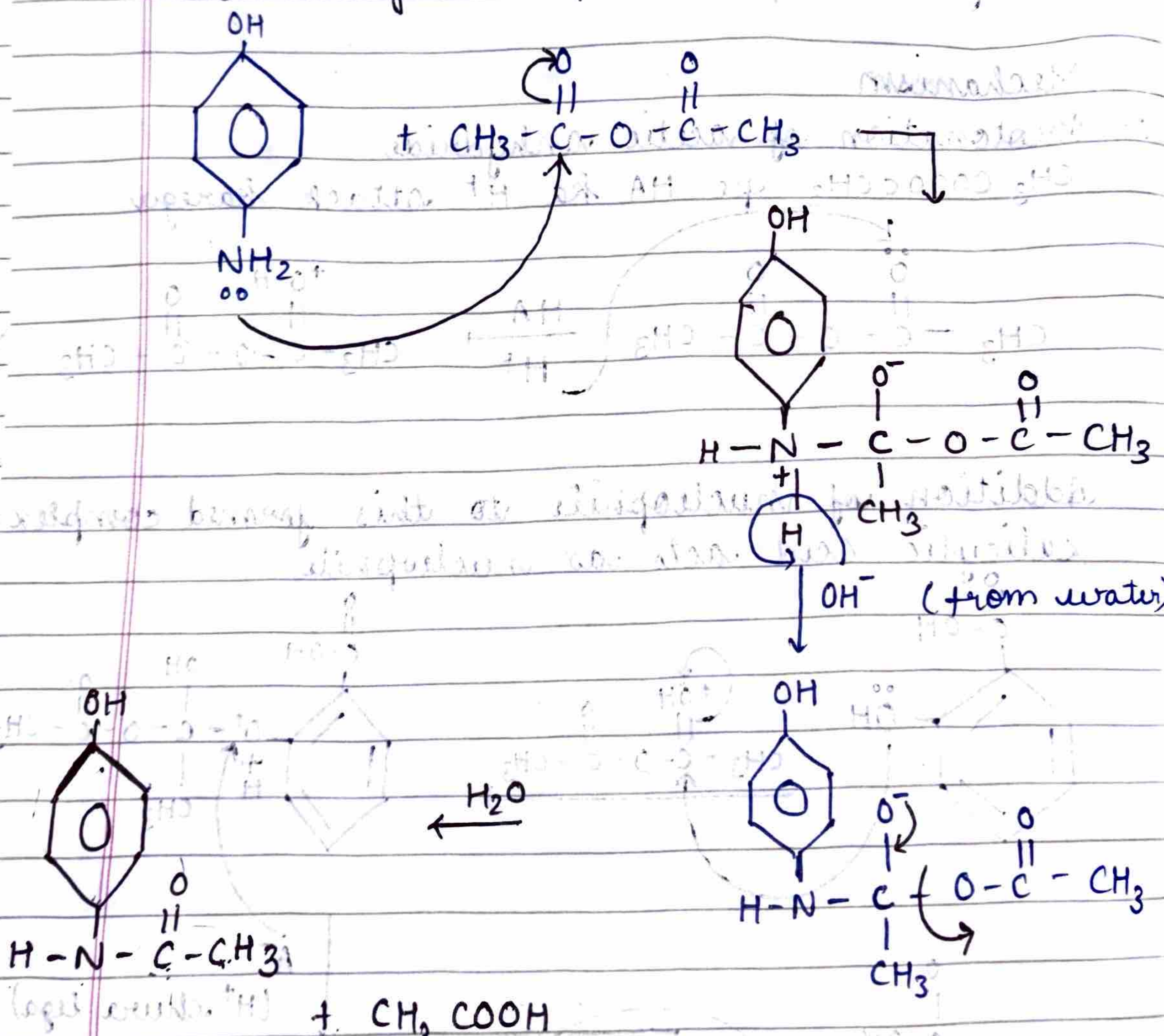
→ Synthesis of Paracetamol → Applications
Fever, Pain relief



• Mechanism

1. Addition of nucleophile 



→ Organic Reactions

1. Elimination (E_1, E_2)
2. Substitution (SN_1, SN_2)
3. Addition & it's types
4. Ring opening
5. ~~SN~~ oxidation, reduction
6. cyclisation

1. Elimination reaction

Two atoms of groups of atoms are removed from adjacent carbon atoms. It is based on saytzeff's rule that alkene with more α H is considered major product

$$CH_3-\underset{\substack{|\\Br}}{CH}-\underset{\substack{|\\H}}{CH}-CH_3 \xrightarrow{NaOH} CH_2=CH-CH_2-CH_3$$

↳ minor (hoffman)

$$CH_3-CH=CH-CH_3$$

↳ major (saytzeff)

- E_1

Leaving group detach to form C^+ & weak base is used to form alkene.

$$CH_3-\overset{\substack{CH_3\\|}}{\underset{\substack{|\\CH_3}}{C}}-OH \xrightarrow{NH_4OH} CH_3-\underset{\substack{|\\CH_3}}{C}=CH_2$$

- E_2

No C^+ formed, Base attract proton, L.G. departs

$$CH_3-CH_2-Br \xrightarrow{KOH} CH_2=CH_2$$

2. Substitution Reaction

An group is added to depart the leaving group

- Nucleophilic substitution
  ↳ proton loving

  Nucleophile comes, leaving group departs

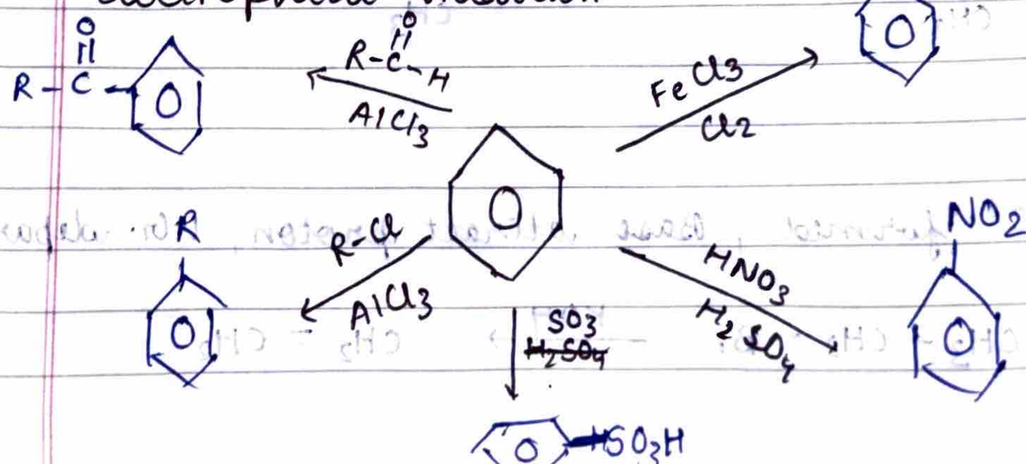
→ SN_2 → strong nucleophile
→ Bimolecular
→ second order
→ Inversion configuration
→ Backside attack, 1 step
→ $1° > 2° > 3°$

$$CH_3Br + OH^- \longrightarrow CH_3OH + Br^-$$

→ SN_1 → strong C^+
→ unimolecular
→ First order
→ Racemic mixture
→ C^+ formation, 2 step
→ $3° > 2° > 1°$

$$CH_3-\underset{CH_3}{\overset{CH_3}{C}}-Br + H_2O \longrightarrow CH_3-\underset{CH_3}{\overset{CH_3}{C}}-OH + HBr$$

- electrophilic Addition



3. Addition Reaction

- electrophilic addition

Addition of H at double / triple bonds

$$CH_3-CH_2=CH_2-CH_3 \xrightarrow{Cl_2} CH_3-\underset{}{\overset{Cl}{CH_2}}-\overset{Cl}{CH_2}-CH_3$$

$$CH_3-CH_2=CH_2-CH_3 \xrightarrow{HCl} CH_3-\underset{Cl}{CH_2}-CH_3-CH_3$$

$$CH_3-CH_2=CH_2-CH_3 \xrightarrow{H_2O} CH_3-\underset{OH}{CH_2}-CH_3=CH_3$$

- Nucleophilic addition

$$R-\overset{O}{\overset{||}{C}}-H + HCN \longrightarrow R-\overset{OH}{\underset{CN}{C}}-H$$

$$R-\overset{O}{\overset{||}{C}}-H + RMgX \longrightarrow R-\overset{OH}{\underset{R}{C}}-H$$

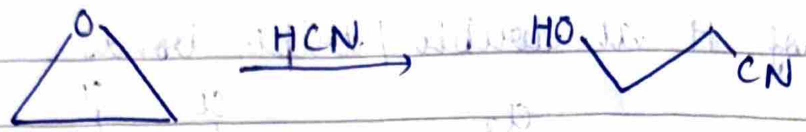
→ Peroxide effect (Khrasch effect)

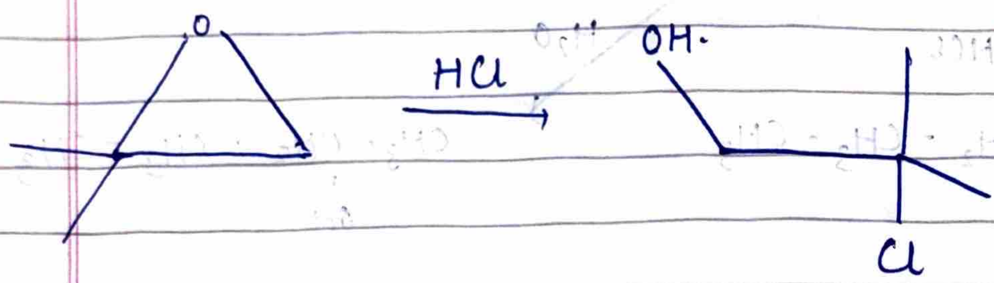
- In presence of peroxide, HBr, addition is done by anti markovnikov's rule. Br^- goes towards less stable C^+
- It follows free radical mechanism

$$CH_2=CH-Cl \xrightarrow[Peroxide]{HBr} Br-CH-CH_2-Cl$$

↓ vinyl chloride

4. Ring opening reaction

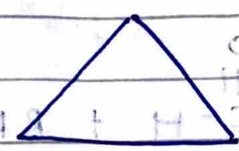




5. Cyclisation Reaction

Diel's Alder Reaction

Butadiene + Ethylene $\longrightarrow$ Cyclohexane

$Cl - CH_2 - CH_2 - CH_2 - Cl \xrightarrow{2Na}$ 

unstable

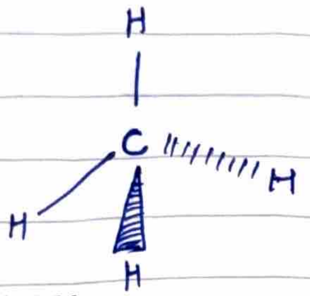
5/6 ki rings $\rightarrow$ stable hoti hai

→ Projections

• wedge - dash

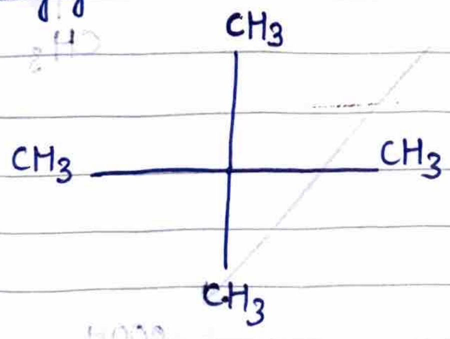
wedge → out of plane

dash → inside plane



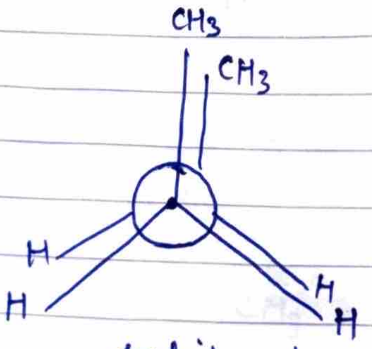
• Fischer

Horizontal line pe wo atom ayege jo out of plane hai aur vertical pe inside plane wale ayege

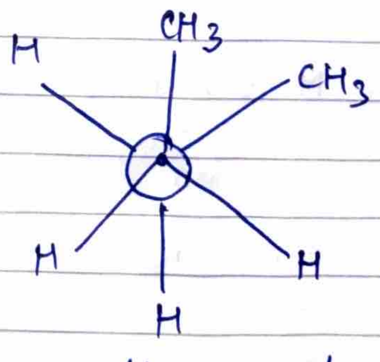


• New mann

Front carbon → (•)   Back carbon → (o)



eclipsed



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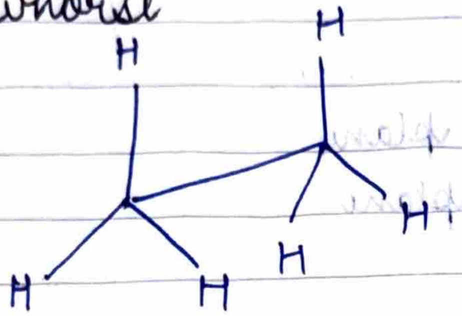
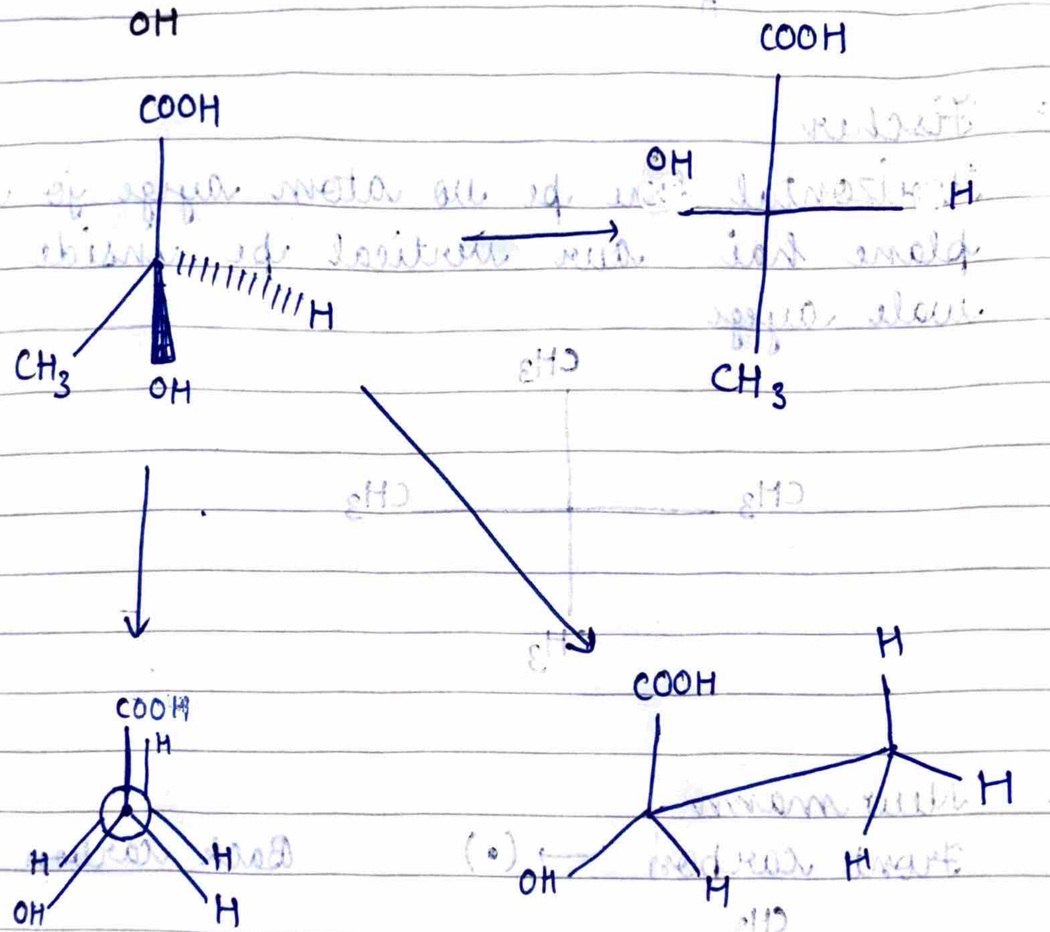
• Sawhorse



eg → $CH_3 - \underset{\underset{OH}{|}}{CH} - COOH$



→ Structural isomers
same molecular formula but diff structural arrangements

- Chain isomer

$C_5 - H_{10}$

C-C-C-C-C   C-C(C)-C-C   C-C(C)(C)-C

- Position isomer

C-C-C-OH   C-C(OH)-C

- Functional isomers
alcohol & ether

C-C-C-C-OH   C-C-O-C-C

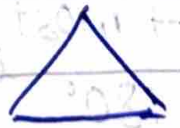
- Metamers
O, ya C=O ya NH ke dono taraf carbon chain present ho toh

C-C-O-C-C-C   C-O-C-C-C-C

- Tautomers
Functional group changes due to H-shift
keto ⇌ enol

$$CH_3 - \overset{O}{\overset{\|}{C}} - CH_3 \longrightarrow CH_3 - \overset{OH}{\overset{|}{C}} = CH_2$$

- Ring chain

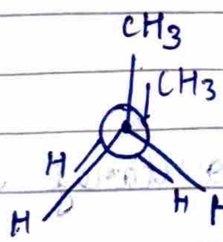
C=C-C   

→ Stereo isomerism
↳ same MF & connections but different arrangements in space.

Conformational	Configurational
• Formed due to rotation about C-C bond	• Bond breaks & forms again
• Less energy gap	• More energy gap
• Instantaneous conversion	• Can't be separated
• Temporary	• Req chem reactions
• unstable	• stable

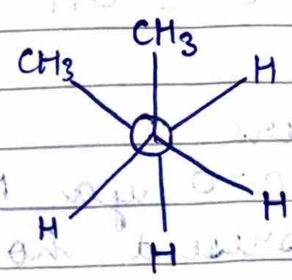
→ conformational isomers
• n - butane

$CH_3 - CH_2 - CH_2 - CH_3$



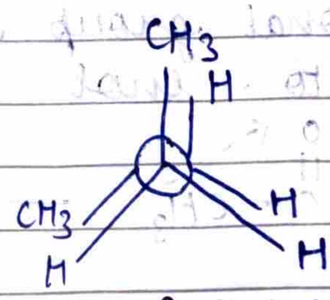
Fully eclipsed
$\theta = 0°$
least stable
most energy

$\xrightarrow{60°}$



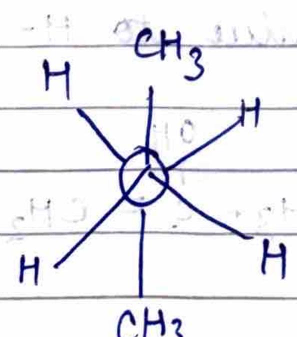
Gauche
$\theta = 60°$

$\downarrow 60°$

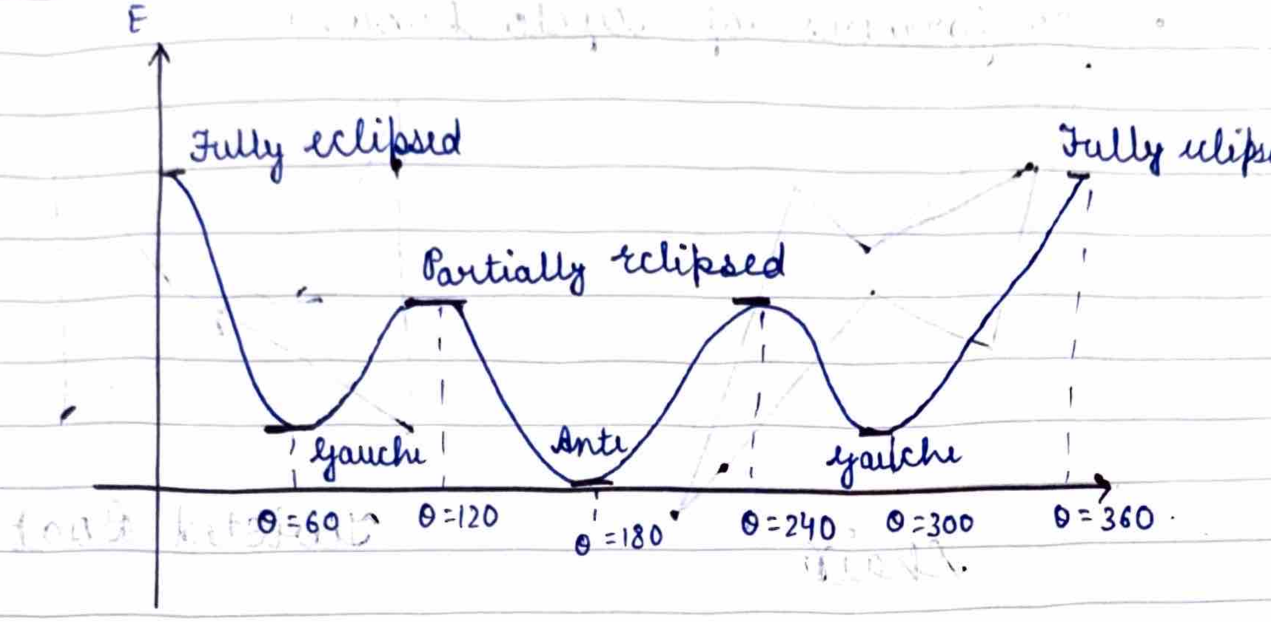


Partially eclipsed
$\theta = 120°$

←

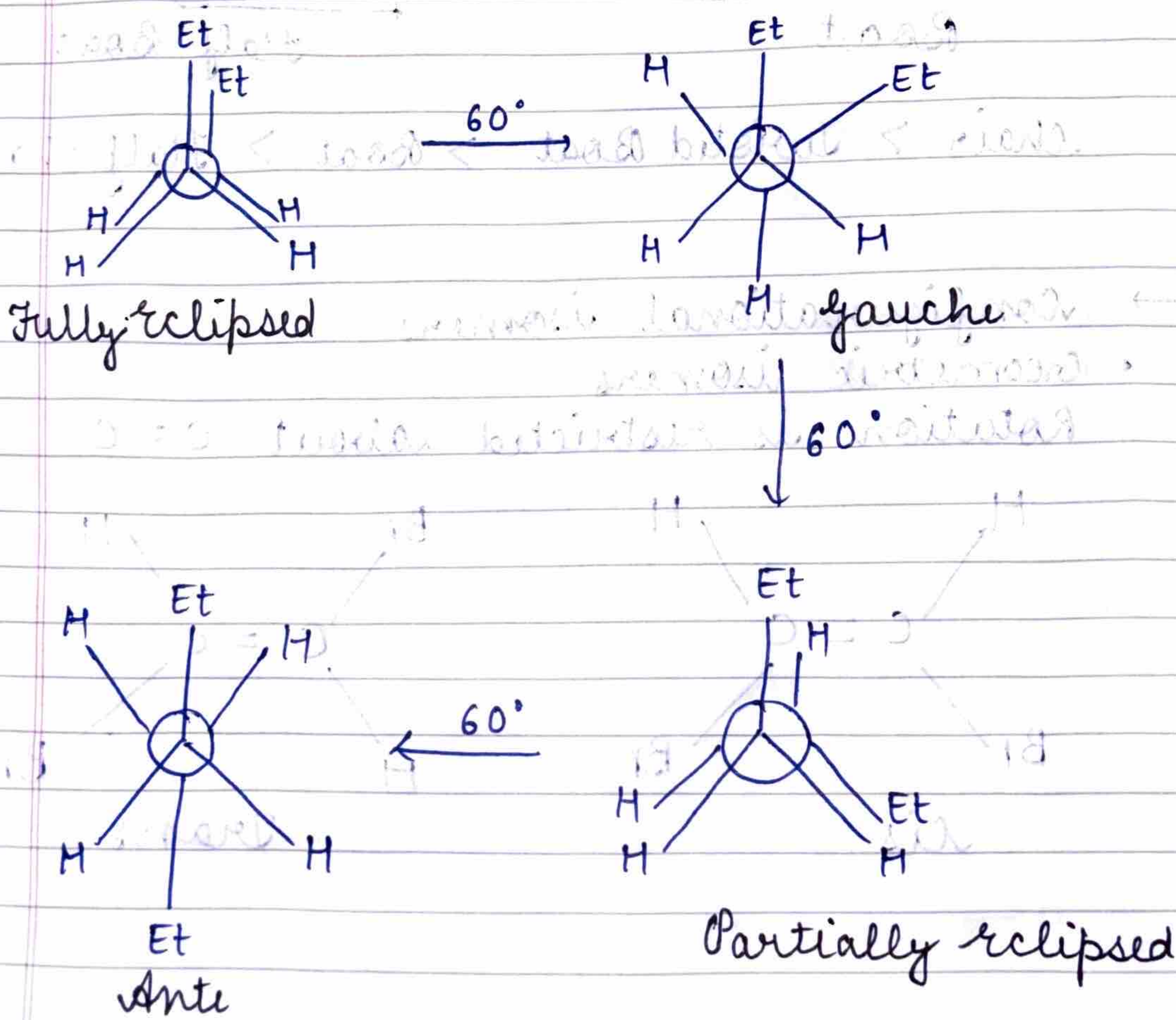


Anti → most stable
least energy
$\theta = 180°$



- n - hexane

$CH_3 - CH_2 - CH_2 - CH_2 \{ CH_2 - CH_3] \rightarrow$ ethyl $\rightarrow$ Et



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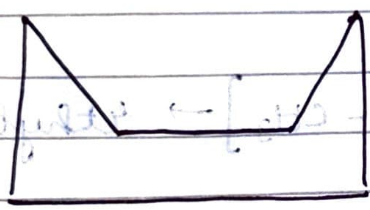
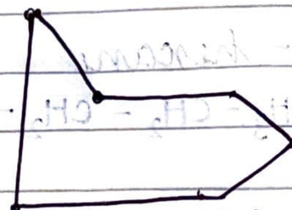
- conformers of cyclo hexane

Chair

Twisted Boat



Boat



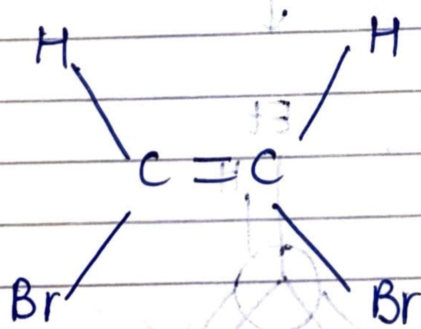
Half Boat

Chair > Twisted Boat > Boat > Half Chair

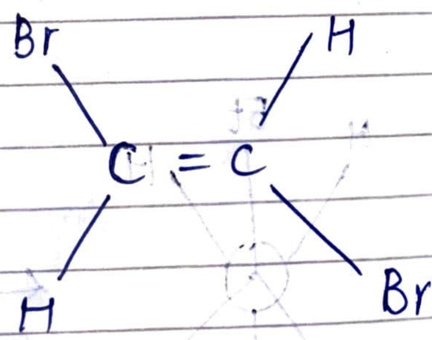
→ configurational isomers

- Geometric isomers

Rotation is restricted about C = C



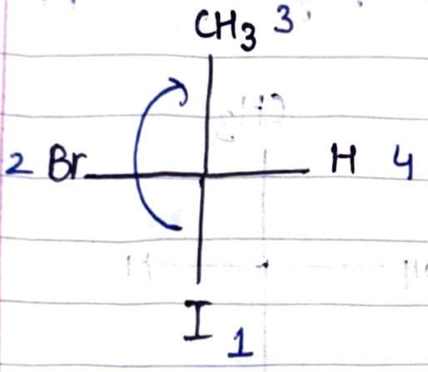
cis

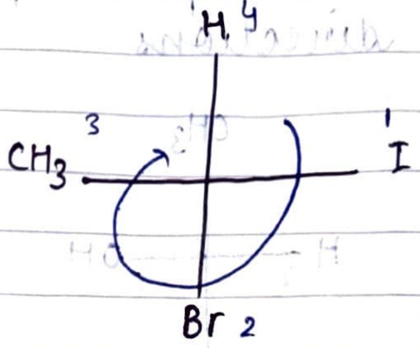


Trans

→ R & S configuration

- Assign priority acc to molar mass
  MM ↑ priority = 1
- see from 4^{th} priority & tell 1 → 2 → 3
  is clock / anticlock
  (R) (S)
- 4^{th} grp horizontal ho to reverse krdo





R hota.
but H on 4^{th}
so it's S

S

→ Chirality

- A molecule is chiral if its non super imposable mirror images of each other
- A chiral center is carbon attached to 4 different groups

$$CH_3 - \overset{*}{C}H(Cl) - CH_2 - CH_3$$

$C^* \rightarrow$ H, Cl, Me, Et
All Four different groups

- eg → lactic acid

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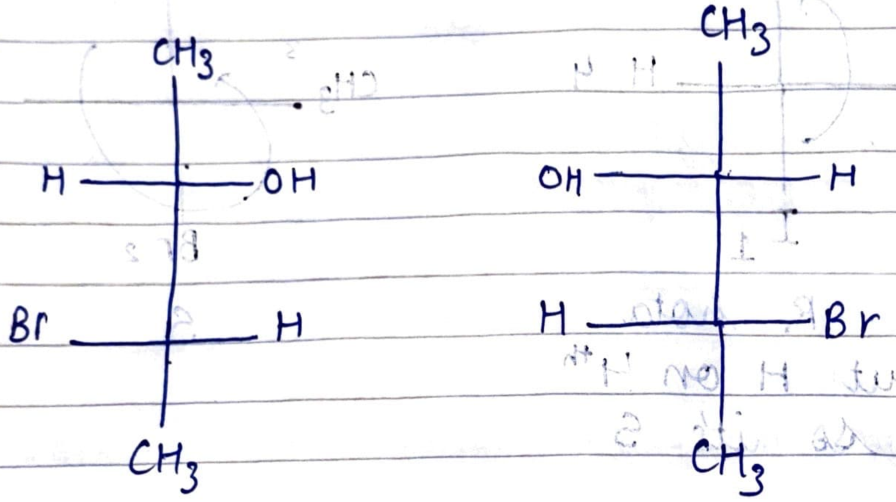
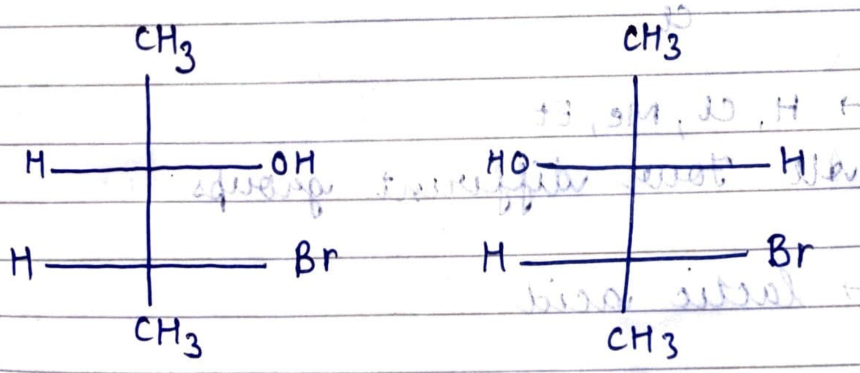
→ Enantiomers

- Pair of stereo isomers that are non super imposable mirror images of each other
- If even 2 C* is present, then it may show enantiomerism
- Always occur in pairs
- Have same physical properties
- Rotate plane polarised light in opposite directions



→ Diasteomers

- Diasteomers are stereoisomers that aren't mirror images of each other
- Present for 2 or more stereo centers
- Different physical properties
- Can be separated by distillation



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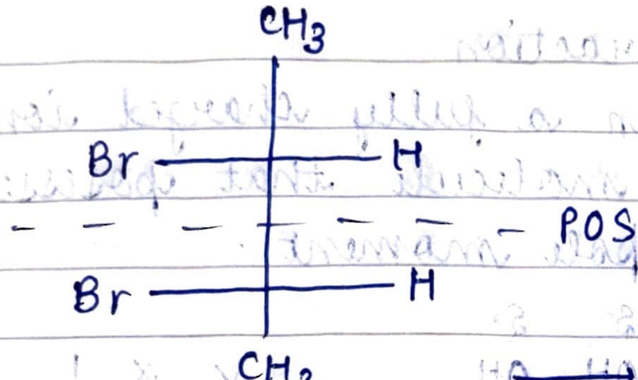
→ Mesomers

- A meso compound is a molecule that has two or more C* but optically inactive due to POS



→ meso compound

→ Optical Activity

- Ability of chiral compound to rotate the plane polarised light
- measured using polarimeter
- If rotates right then dextro-rotatory
- If rotates left then levo-rotatory
- Present in unsymmetric compounds